
Stress-Testing Wanda: A Reproducibility and Robustness Study of Activation-Based LLM Pruning

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Abstract

1 We present a systematic stress-test of Wanda [?], a simple and effective pruning
2 method for large language models (LLMs) that scores weights by the product of
3 their magnitude and the ℓ_2 -norm of corresponding input activations. The original
4 paper claims robustness to the choice of calibration data, but tests this only by
5 varying the *quantity* of English C4 samples—never the *domain* or *language*. We
6 argue that the activation norm $\|X_j\|_2$ is a statistical fingerprint of the calibration
7 distribution, and that domain-mismatched calibration can make Wanda perform
8 *worse* than the calibration-free magnitude pruning baseline on capabilities not
9 represented in the calibration data. We test this thesis on LLaMA across three
10 experiments: (1) calibration domain shift at 50% sparsity using English (C4),
11 Korean (MC4-ko), code (The Stack), and random-token calibrations; (2) per-task
12 fragility curves across 50%, 60%, and 70% sparsity; and (3) the calibration $\times$
13 sparsity interaction at 70% sparsity. We additionally document reproducibility
14 issues encountered when running the original codebase under modern library
15 versions, and provide an English-only domain-shift supplement on LLaMA-2-7B
16 to isolate the cross-lingual component of the effect.

17 1 Introduction

18 Model compression is a critical enabler of efficient inference for large-scale neural networks. Among
19 compression techniques, unstructured pruning offers a flexible way to reduce model size by zeroing
20 out individual weights. Wanda [?] (Pruning by **W**eights **a**nd **A**ctivations) proposes a remarkably
21 simple pruning criterion: for each weight w_{ij} , its importance score is defined as

$$S_{ij} = |w_{ij}| \cdot \|\mathbf{x}_j\|_2, \quad (1)$$

22 where $\mathbf{x}_j$ denotes the j -th input feature activation collected over a small calibration set. Weights with
23 the lowest scores are pruned. This requires no gradient computation and runs in minutes even for
24 65B-parameter models, making it highly practical.

25 However, the original evaluation tests calibration-data robustness only by varying the *quantity* of
26 C4 [?] samples in a single domain (English web text). This leaves a sharper question untested: *can*
27 *domain-mismatched calibration actively harm Wanda’s pruning quality, to the point of being worse*
28 *than calibration-free magnitude pruning?* We focus on this single thesis through three experiments:

- 29 • **Calibration domain shift.** At 50% sparsity, does using English, Korean, code, or random-
30 token calibration affect which capabilities (English knowledge, math reasoning, Korean
31 language) survive pruning?

- **Per-task fragility curves.** As sparsity increases from 50% to 70%, do different downstream tasks degrade at different thresholds, and does perplexity track them?
- **Calibration $\times$ sparsity interaction.** At the most aggressive setting, do calibration-domain damage and sparsity-cliff damage compound super-additively, suggesting both effects target the same low-activation features?

Following the Track 2 stress-testing protocol, we first reproduce the original codebase (and document the reproducibility issues encountered), then run the three experiments above on LLaMA, with a LLaMA-2-7B English-only supplement to isolate the cross-lingual component.

2 Background

2.1 Wanda Pruning

Given a pre-trained LLM, Wanda prunes each linear layer independently by computing importance scores S_{ij} for all weights and zeroing out those below a target sparsity ratio s . The key insight over magnitude-only pruning is that a weight connected to a high-activation feature should be treated as more important, even if its magnitude is small. The calibration set is used solely for a single forward pass to collect activation statistics; no weight updates are performed.

2.2 Baselines

We compare Wanda against:

- **Magnitude pruning:** scores weights by $|w_{ij}|$ only.
- **SparseGPT [?]:** solves a layer-wise reconstruction problem; more expensive but generally stronger.
- **Dense baseline:** unpruned model performance.

3 Reproducibility Issues

Before conducting stress-tests, we attempted to reproduce the original results using the public codebase.¹ We identified the following compatibility issues with modern library versions (torch 2.6, transformers 5.5, datasets 3.x):

1. **torch < 2.6 blocked:** A security vulnerability (CVE-2025-32434) in `torch.load` causes the `transformers` library to raise a `ValueError` when loading model weights with torch versions below 2.6. The original codebase was developed with torch 2.x, requiring an upgrade.
2. **C4 dataset config renamed:** The `datasets` library changed the C4 configuration identifier from `'allenai-c4'` to `'en'`, causing a `BuilderConfig` error at runtime.
3. **Split size verification:** Even after the config fix, loading only a subset of C4 (one shard) triggers a `NonMatchingSplitsSizesError`. The fix requires passing `verification_mode='no_checks'`.
4. **OPT architecture mismatch:** Wanda's calibration code accesses `model.model.layers`, which is valid for LLaMA but fails for OPT, whose layers reside at `model.model.decoder.layers`.

These issues collectively demonstrate that the original codebase lacks robustness to software environment changes, raising reproducibility concerns. All fixes were applied to our fork before proceeding with experiments.

¹<https://github.com/locuslab/wanda>

72 4 Experimental Setup

73 4.1 Compute Environment

74 All experiments are conducted on a server equipped with $8 \times$ NVIDIA RTX 6000 Ada Generation
75 GPUs (48 GB VRAM each), running CUDA 12.6, PyTorch 2.6.0, and transformers 5.5.3.

76 4.2 Models

- 77 • **LLaMA-3-8B** [?]: main model. Chosen because it scores above random chance on Korean
78 benchmarks (KMMLU, KoBEST), which is required to measure Korean-capability loss
79 after pruning.
- 80 • **LLaMA-2-7B** [?]: supplementary model for the English-only domain-shift control (Sec-
81 tion ??).

82 4.3 Calibration Datasets

83 We use 128 samples per source, matching the original Wanda setup:

- 84 • **C4** [?] (default): English web text.
- 85 • **MC4-ko**: Korean subset of the multilingual C4.
- 86 • **The Stack**: source-code corpus (a non-natural-language domain).
- 87 • **Random tokens**: tokens sampled uniformly from the vocabulary; a structureless control.

88 4.4 Evaluation Metrics

89 All metrics are computed with the `lm-evaluation-harness` framework where applicable.

- 90 • **WikiText-2 perplexity** ($\downarrow$): primary metric in the original Wanda paper; English language-
91 modeling fluency.
- 92 • **MMLU** ($\uparrow$, 5-shot accuracy): English world-knowledge and reasoning.
- 93 • **GSM8K** ($\uparrow$, 8-shot accuracy): English mathematical reasoning.
- 94 • **KoBEST HellaSwag** ($\uparrow$, 0-shot accuracy): Korean commonsense reasoning—the capability
95 we expect to be most sensitive to English-only calibration.

96 5 Results

97 5.1 Experiment 1: Calibration Domain Shift at 50% Sparsity

98 We hold model (LLaMA-3-8B) and sparsity (50%, unstructured) fixed and vary the calibration source
99 for Wanda, comparing against two anchor baselines that do not depend on calibration data the same
100 way. Table ?? reports six pruning runs.

101 5.2 Experiment 2: Per-Task Fragility Curves Across Sparsity

102 We hold calibration fixed (C4) and sweep sparsity at 50%, 60%, and 70%. The 50% row for each
103 method is shared with Table ??, so this table contributes six *new* pruning runs. The goal is to identify
104 whether different downstream tasks cliff-drop at different sparsity thresholds, and whether perplexity
105 tracks those drops.

106 5.3 Experiment 3: Calibration $\times$ Sparsity Interaction

107 We test the extreme corner of the design space: English-calibrated and Korean-calibrated Wanda,
108 both at 50% and 70% sparsity. Three of the four cells are reused from Tables ?? and ??; only *Wanda*
109 (*MC4-ko*) at 70% is a new run. If the Korean-capability loss from English calibration and from high
110 sparsity compound super-additively on KoBEST HellaSwag, this would suggest both effects target
111 the same low-activation Korean features.

Table 1: Effect of calibration source on Wanda at 50% unstructured sparsity (LLaMA-3-8B). The activation norm $\|X_j\|_2$ is a statistical fingerprint of the calibration distribution, so we expect domain-mismatched calibration to disproportionately damage capabilities not represented in that distribution. **[TO BE FILLED]**

Method (calibration)	WT-2 ppl ↓	MMLU ↑	GSM8K ↑	KoBEST HS ↑
Dense (no pruning)	—	—	—	—
Magnitude (no calibration)	—	—	—	—
SparseGPT (C4)	—	—	—	—
Wanda (C4, English web)	—	—	—	—
Wanda (MC4-ko, Korean)	—	—	—	—
Wanda (The Stack, code)	—	—	—	—
Wanda (random tokens)	—	—	—	—

Table 2: Per-task fragility under increasing sparsity, LLaMA-3-8B with C4 calibration where applicable. **[TO BE FILLED]**

Method	Sparsity	WT-2 ppl ↓	MMLU ↑	GSM8K ↑	KoBEST HS ↑
Dense	0%	—	—	—	—
Magnitude	50%	—	—	—	—
Magnitude	60%	—	—	—	—
Magnitude	70%	—	—	—	—
SparseGPT (C4)	50%	—	—	—	—
SparseGPT (C4)	60%	—	—	—	—
SparseGPT (C4)	70%	—	—	—	—
Wanda (C4)	50%	—	—	—	—
Wanda (C4)	60%	—	—	—	—
Wanda (C4)	70%	—	—	—	—

112 5.4 Experiment 4 (Supplement): English-Only Domain Shift on LLaMA-2-7B

113 To isolate the cross-lingual component of Experiment 1, we run a parallel domain-shift study on
 114 LLaMA-2-7B—which has limited Korean capability—using only English-domain calibrations. If
 115 within-English domain shift (web text vs. code vs. random tokens) produces *less* damage than the
 116 cross-lingual shift in Experiment 1, that supports the thesis that the language axis dominates the
 117 domain axis for Wanda.

118 **Run accounting.** Tables ??–?? together cover $6 + 6 + 1 + 13 = 26$ distinct pruning runs; the
 119 dense rows do not count as pruning runs. Cells reused across tables (e.g., Wanda(C4) at 50% in both
 120 Tables ?? and ??) are counted once.

121 6 Analysis

122 6.1 Calibration Domain Sensitivity

123 **[To be written after experiments.]** We hypothesize that English-calibrated Wanda will damage
 124 Korean capabilities (KoBEST HellaSwag) more than Korean-calibrated Wanda damages English
 125 capabilities (MMLU, GSM8K), because Korean text naturally contains some English tokens but
 126 English text contains essentially no Korean. The decisive prediction is that English-calibrated Wanda
 127 on KoBEST HellaSwag falls *at or below* the calibration-free magnitude pruning baseline—if so,
 128 calibration data has been actively counterproductive.

129 6.2 Sparsity Cliff and Per-Task Fragility

130 **[To be written after experiments.]** We expect WikiText-2 perplexity to degrade approximately
 131 continuously, while downstream tasks cliff-drop at distinct thresholds—likely GSM8K and KoBEST

Table 3: Compound effect of calibration mismatch and high sparsity on Wanda (LLaMA-3-8B). Δ columns report drop relative to the dense baseline (filled after results are available). **[TO BE FILLED]**

Calibration	Sparsity	MMLU $\uparrow$	Δ MMLU	KoBEST HS $\uparrow$	Δ KoBEST
– (Dense)	0%	–	0	–	0
C4 (English)	50%	–	–	–	–
C4 (English)	70%	–	–	–	–
MC4-ko (Korean)	50%	–	–	–	–
MC4-ko (Korean)	70%	–	–	–	–

Table 4: English-only domain shift on LLaMA-2-7B. All evaluations are on English benchmarks; Magnitude and SparseGPT are reported only at the two corner sparsities for compute reasons. **[TO BE FILLED]**

Method (calibration)	Sparsity	WT-2 ppl $\downarrow$	MMLU $\uparrow$	ARC-c $\uparrow$
Dense	0%	–	–	–
Magnitude	50%	–	–	–
Magnitude	70%	–	–	–
SparseGPT (C4)	50%	–	–	–
SparseGPT (C4)	70%	–	–	–
Wanda (C4, web)	50%	–	–	–
Wanda (C4, web)	60%	–	–	–
Wanda (C4, web)	70%	–	–	–
Wanda (The Stack)	50%	–	–	–
Wanda (The Stack)	60%	–	–	–
Wanda (The Stack)	70%	–	–	–
Wanda (random tokens)	50%	–	–	–
Wanda (random tokens)	60%	–	–	–
Wanda (random tokens)	70%	–	–	–

132 HellaSwag before MMLU. This would imply that perplexity is a poor proxy for downstream capability
133 retention under aggressive pruning.

134 6.3 Calibration $\times$ Sparsity Interaction

135 **[To be written after experiments.]** If the Korean-capability loss from English calibration (Ex-
136 periment 1) and from high sparsity (Experiment 2) compound super-additively at 70% sparsity
137 on KoBEST HellaSwag, this would suggest both effects damage the same set of low-activation
138 Korean-specific features, rather than two independent failure modes.

139 6.4 Cross-Lingual vs. Within-English Domain Shift

140 **[To be written after experiments.]** The LLaMA-2-7B supplement (Section ??) lets us compare
141 the within-English domain shift (web text $\rightarrow$ code $\rightarrow$ random tokens) against the cross-lingual shift
142 (English $\rightarrow$ Korean) from Experiment 1. We expect the cross-lingual shift to produce strictly more
143 damage on the affected language’s benchmarks, since dormant features are fully missed rather than
144 merely under-weighted.

145 7 Limitations

146 Our study has the following limitations:

- 147 • **Scope:** Due to compute constraints, we do not evaluate LLaMA-30B or larger models.
- 148 • **Rigor:** Each experiment is run with a single random seed; variance across seeds is not
149 reported.

- 150 • **Baselines:** We do not compare against more recent pruning methods (e.g., RIA [?]) due to
151 time constraints.
- 152 • **Hardware comparison:** Hardware-level benchmarks (latency, throughput) for actual sparse
153 inference are not included, as unstructured sparsity does not directly translate to speedup on
154 standard GPUs without specialized kernels.

155 8 Conclusion

156 We conducted a systematic stress-test of Wanda, a simple yet effective LLM pruning method, along
157 four axes: model architecture, calibration data, sparsity ratio, and evaluation metrics. We additionally
158 documented four reproducibility issues in the original codebase arising from library version updates.
159 **[Conclusions to be updated after experiments.]** Our results suggest that while Wanda is a strong
160 baseline under its original evaluation conditions, its performance is sensitive to **[X, Y, Z]**, which
161 should be considered when applying the method in practice.

162 References

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173 **NeurIPS Paper Checklist**

174 **1. Claims**

175 Answer: [Yes]

176 Justification: The abstract and introduction clearly state that this is a stress-testing study of
177 Wanda, not a proposal of a new method. Claims are scoped accordingly.

178 **2. Limitations**

179 Answer: [Yes]

180 Justification: Section 6 explicitly discusses limitations in scope, rigor, baselines, and hard-
181 ware evaluation.

182 **3. Theory assumptions and proofs**

183 Answer: [N/A]

184 Justification: This paper does not include theoretical results.

185 **4. Experimental result reproducibility**

186 Answer: [Yes]

187 Justification: All commands, library versions, and hardware configurations are documented.
188 Our modified code is publicly available.

189 **5. Open access to data and code**

190 Answer: [Yes]

191 Justification: Our fork with all modifications is available at [https://github.com/](https://github.com/[your-account]/wanda)
192 [\[your-account\]/wanda](https://github.com/[your-account]/wanda).

193 **6. Experimental setting/details**

194 Answer: [Yes]

195 Justification: Section 4 describes all models, datasets, hyperparameters, and evaluation
196 metrics.

197 **7. Experiment statistical significance**

198 Answer: [No]

199 Justification: Due to compute constraints, experiments are run with a single seed. Error bars
200 are not reported.

201 **8. Experiments compute resources**

202 Answer: [Yes]

203 Justification: Section 4.1 reports the GPU type, VRAM, CUDA version, and software stack
204 used for all experiments.

205 **9. Code of ethics**

206 Answer: [Yes]

207 Justification: This work studies model compression of publicly available models and poses
208 no ethical concerns.

209 **10. Broader impacts**

210 Answer: [N/A]

211 Justification: This is a reproducibility study with no direct negative societal impact.

212 **11. Safeguards**

213 Answer: [N/A]

214 Justification: We do not release new models or datasets.

215 **12. Licenses for existing assets**

216 Answer: [Yes]

217 Justification: Wanda (MIT License), OPT (OPT-175B License), LLaMA-2 (LLaMA 2
218 Community License), C4 (ODC-BY) are all properly cited.

219 **13. New assets**
220 Answer: [N/A]
221 Justification: No new datasets or models are introduced.
222 **14. Crowdsourcing and research with human subjects**
223 Answer: [N/A]
224 Justification: No human subjects are involved.
225 **15. Institutional review board (IRB) approvals**
226 Answer: [N/A]
227 Justification: No human subjects are involved.
228 **16. Declaration of LLM usage**
229 Answer: [N/A]
230 Justification: LLMs are the *subject* of study, not a component of the methodology.